

Homework 3.2

Problem 1

Find the smallest positive starting point for which Newton's method diverges when it is applied to $f(x) = \tan^{-1}(x)$.

Since we are looking for the value where Newton's method will not converge, we will look for the case where $x_{n+1} = -x_n$ since this is where we would enter a cycle.

$$f(x) = \tan^{-1}(x), \frac{df}{dx} = \frac{1}{1+x^2}$$

$$x_{n+1} = x_n - \tan^{-1}(x) (1+x^2)$$

$$-x = x - \tan^{-1}(x) (1+x^2) \Rightarrow 0 = 2x - \tan^{-1}(x) (1+x^2)$$

Using Newton's Method (Computer Problem 1) with initial guess $x_0 = 1$ give us that:

$$g(x) = 2x - \tan^{-1}(x) (1+x^2)$$

$$\frac{dg}{dx} = 2 - 1 - 2x * \tan^{-1}(x) = 1 - 2x * \tan^{-1}(x)$$

```
>> N(g,dg,1)
ans = 1.391745200270735
```

Now doing the following:

```
>> ep = 10^(-8)
ep = 1.0000000000000000e-08
>> fail = N(g,dg,1)
fail = 1.391745200270735
>> ep = 10^(-8);
>> N(f,df,fail-ep)
ans = 0
>> N(f,df,fail+ep)
ans = NaN
```

Thus, we can see that this point is where the function fails to converge.

Problem 5

What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2 y, y \in \mathbb{R}$$

Identify it as the newton iteration for a certain function.

$$x_{n+1} = x_n - (-x_n + x_n^2 y)$$

$$\frac{f}{f'} = x^2 y - x \Rightarrow f' = \frac{f}{x(xy - 1)} = f \left[\frac{A}{x} + \frac{B}{xy - 1} \right]$$

$$A(xy - 1) + Bx = 1 \Rightarrow A = -1, -xy + Bx = 0 \Rightarrow B = y$$

$$\frac{df}{dx} f' = f \left[\frac{y}{xy - 1} - \frac{1}{x} \right]$$

$$\frac{df}{f} = \left[\frac{y}{xy - 1} - \frac{1}{x} \right] dx$$

$$\ln(f) = \int \frac{y}{xy - 1} dx - \int \frac{dx}{x}$$

$$\ln(f) = \ln(xy - 1) - \ln(x) = \ln\left(\frac{xy - 1}{x}\right)$$

$$\therefore f(x) = \frac{xy - 1}{x}$$

Now to verify:

$$f'(x) = \frac{x * (y) - (xy - 1)(1)}{x^2} = \frac{1}{x^2}$$

$$\frac{f(x)}{f'(x)} = \frac{xy - 1}{x} * \frac{x^2}{1} = x(xy - 1)$$

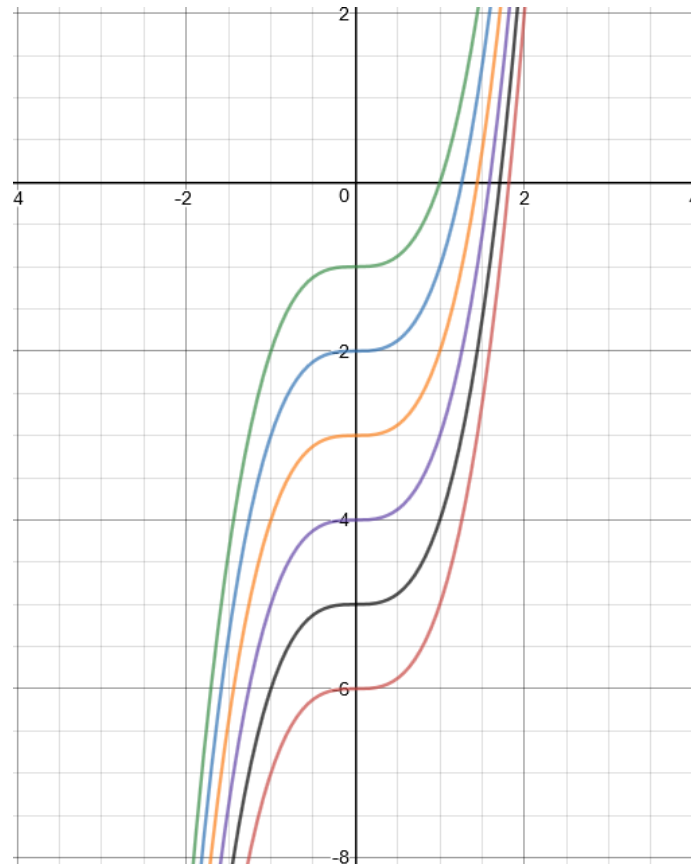
$$\therefore x_{n+1} = x_n - (x_n^2 y - x_n) = 2x_n - x_n^2 y$$

Problem 10

Derive a Newton iteration formula for computing $\sqrt[3]{R}$ where $R > 0$. Perform a graphical analysis of your function $f(x)$ to determine the starting values for which the iteration will converge.

$$\sqrt[3]{R} \Rightarrow x^3 - R = 0 \Rightarrow f(x) = x^3 - R, \frac{df}{dx} = 3x^2$$

$$x_{n+1} = x_n - \frac{x_n^3 - R}{3x_n^2} = x_n - \frac{x_n}{3} - \frac{R}{3x_n^2} = \frac{1}{3} \left(2x_n - \frac{R}{x_n^2} \right)$$



Since we see that the function $x^3 - R$ is convex for all values greater than $x = 0$, we see that the function will converge. Letting $h = x^3 - 3$ and $\frac{dh}{dx} = 3x^2$, we have that:

```
>> N(h,dh,10)
ans = 1.442249570307408
>> N(h,dh,.1)
ans = 1.442249570307408
>> N(h,dh,500)
ans = 1.442249570307408
>> N(h,dh,10^8)
ans = 1.442249570307408
>> N(h,dh,10^-8)
ans = 1.442249570307408
>> N(h,dh,0)
ans = NaN
```

Problem 15

Consider a variation of Newton's in which only one derivative is needed; that is,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$$

Find C and s such that

$$e_{n+1} = C e_n^s$$

Doing what we did to find the convergence of the original Newton's method, and defining $e_0 = x_0 - r$, and know that our guess was relatively close:

$$r + e_{n+1} = r + e_n - \frac{f(r) + e_n f'(r) + \dots}{f'(r) + f''(r)e_0 + \dots} \Rightarrow e_{n+1} = e_n - \frac{e_n f'(r) + e_n^2 f''(r) + O(h^3)}{f'(r) + f''(r)e_0 + O(h^3)}$$

$$e_{n+1} = e_n - e_n \frac{(f'(r) + e_n f''(r))}{f'(r) + e_0 f''(r)}$$

...

[Did not finish]

Computer Problems

Problem 5

The equation $2x^4 + 24x^3 + 61x^2 - 16x + 1 = 0$ has two roots near $x = 0.1$. Determine them by means of Newton's Method.

Newton's Method

```
function Newton = N(f,df,x)
%This function will take in the function in question, its derivative, and
%the initial guess of the root and output the root within a certain margin
%as long as the initial guess is close.

delta = 10^(-8); %Our tolerance.
M = 10^3; %Maximum number of iterations.
N = 1; %Initialization.

while ((abs(f(x)) > delta) || (N < M)) %Will repeat the process until
%either we are within our tolerance or we have exceeded our steps.
    x = x - f(x)/df(x); %Algorithm.
    N = N + 1; %Increase count.
end

x; %Our final root value.
```

$$f(x) = 2x^4 + 24x^3 + 61x^2 - 16x + 1$$

$$f'(x) = 8x^3 + 72x^2 + 122x - 16$$

Using the value of $x = .1$, we get that:

```
>> f = @(x) 2*x^4+24*x^3+61*x^2-16*x+1;  
>> df = @(x) 8*x^3+72*x^2+122*x-16;  
>> format long  
>> root = N(f,df,0.1)  
root = 0.121320343559643  
>> f(root)  
ans = 0
```

Now since we know that there is another root near .1, we can consider two points, $x = 0$ and $x = 0.2$.

Using $x = 0$:

```
>> root = N(f,df,0)  
root = 0.121320343559643
```

Which is the same root we got before. Now for $x = 0.2$:

```
>> root = N(f,df,0.2)  
root = 0.123105625617660  
>> f(root)  
ans = 0
```

Now we see that this root is different from what we got originally.

Thus, our two roots near 0.1 are:

$$r_1 = 0.121320343559643, r_2 = 0.123105625617660$$

Problem 14

Using Newton's Method, find the roots of the nonlinear system

$$\begin{cases} 4y^2 + 4y + 52x = 19 \\ 169x^2 + 3y^2 + 111x - 10y = 10 \end{cases}$$

The two functions we are trying to find the root for is:

$$f_1(x, y) = 4y^2 + 4y + 52x - 19$$

$$f_2(x, y) = 169x^2 + 3y^2 + 111x - 10y - 10$$

The Jacobian matrix is given by:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$$

$$\frac{\partial f_1}{\partial x} = 52, \frac{\partial f_1}{\partial y} = 8y + 4$$

$$\frac{\partial f_2}{\partial x} = 338x + 111, \frac{\partial f_2}{\partial y} = 6y - 10$$

Now for the Newton's Method for systems:

$$X = \begin{bmatrix} x \\ y \end{bmatrix}, F = \begin{bmatrix} f_1(x, y) \\ f_2(x, y) \end{bmatrix} = \begin{bmatrix} 4y^2 + 4y + 52x - 19 \\ 169x^2 + 3y^2 + 111x - 10y - 10 \end{bmatrix}$$

$$J(X) = F'(X) = \begin{bmatrix} 52 & 8y + 4 \\ 338x + 111 & 6y - 10 \end{bmatrix}$$

$$H = -J(X)^{-1} * F(X)$$

$$X^{(k+1)} = X^{(k)} + H^{(k)}$$

Newton's Method (System)

```
function NewtonSystem = NS(F,DF,X) %This function will take in the function
%as a vector valued function and its derivative (Jacobian) as well as an
%intial guess and from there it will iterate until the norm of the function
%at the X^(kth) iteration is within range or when the maximum number of
%steps have been reached.

delta = 10^-(8); %Our tolerance.
M = 10^3; %Maximum number of iterations.
N = 1; %Initialized value.

while (norm(F(X))>delta) || (N < M) %This loop will keep iterating until
%either the norm of the function is within range or the number of
%iterations have maxed out.
    X = X - (DF(X)^(-1))*F(X); %Newton's Method.
    N = N + 1; %Increase count.
end

NewtonSystem = X; %Our final root value.
```

I will be using the initial guess of $X = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and see if it reaches a root:

```
>> F = @(X) [4*X(2)^2+4*X(2)+52*X(1)-19;169*X(1)^2+3*X(2)^2+111*X(1)-10*X(2)-10];
>> DF = @(X) [52,8*X(2)+4;338*X(1)+111,6*X(2)-10];
>> X = [0;0];
```

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```
>> Root = NS(F,DF,X)
```

```
Root =
```

```
0.134197052397750
```

```
1.304283325542098
```

```
>> F(Root)
```

```
ans = 1.0e-14 *
```

```
0
```

```
-0.177635683940025
```

```
>> norm(F(Root))
```

```
ans = 1.776356839400251e-15
```

As we can see, the norm of our function evaluated at our root is very small thus we have found a suitable root for our problem.